Conflict-Aware Retrieval-Augmented Generation for Multi-Objective Planning Domains

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Abstract— **Standard answers from large language models (LLMs), used with zero-shot or standard retrieval-augmented generation (RAG), often fail to understand the constraints and resolve the inherent conflicting requirements present in real-world planning. This work introduces Conflict-Aware Retrieval-Augmented Generation (CA-RAG), a RAG framework that combines few-shot learning with RAG to provide LLM based planning context for resolving real-world conflicts and provide a reasonable baseline for solution tradeoffs. The system performs LLM-based constraint extraction, rule-based conflict detection, conflict-aware retrieval, score the examples using a hybrid Jaccard + semantic scoring function, create a structured few-shot prompt construction and response generation via Llama-3.3-70B on the Groq API and then evaluate the response based on it's ability to satisfy real-world constraints, resolve conflict and the feasibility of the provided solution. The system is evaluated across three planning domains- health, academic, and travel using 66 annotated cases and 15 held-out queries. CA-RAG achieves a Conflict Resolution Quality (CRQ) of 0.671 — a 117.6% improvement over semantic RAG (0.308) and an overall score of 0.773, versus 0.612 for zero-shot (p = 0.006, Cohen's d = 2.39). Cross-domain retrieval occurs in 22.4% of cases, indicating that conflict structure is the driving force behind useful retrieval.**

Keywords—
Retrieval-Augmented Generation, Constraint Satisfaction, Conflict Detection, Few-Shot Learning, Large Language Models, Multi-Objective Planning

I. INTRODUCTION

Planning in the real world involves various negotiable and non-negotiable constraints and goals that may mutually conflict. For example, a vegetarian student with a daily food budget of INR 100, no kitchen access and a goal to gain muscles, presents a conflict between constraints and goals that get ignored by the automated planning assistance systems. The goal (muscle gain) may be possible but is difficult given these constraints. Standard LLMs suggest non-vegetarian proteins, costly supplements, or meal-prep plans that are useless for the user. Retrieval-Augmented Generation (RAG) helps by grounding generation in examples, but conventional RAG retrieves on lexical or semantic similarity, not conflict structure. This work proposes Conflict-Aware RAG (CA-RAG) to close this gap.

CA-RAG classifies the constraints in two categories- (a) hard constraints or non-negotiable user requirements and (b) soft constraints or tradeoffs. Given a query Q with hard constraints C_hard, soft constraints C_soft and goals G that cannot all be simultaneously satisfied, the problem is to (a) identify the conflict structure of Q, (b) retrieve resolved cases with similar conflict structure, and (c) generate a response that explicitly names conflicts, proposes concrete trade-offs, and satisfies all hard constraints to the extent feasible.

Key contributions:

1. Conflict-Aware Retrieval Algorithm: A hybrid conflict-aware retrieval algorithm combining conflict-type Jaccard overlaps with sentence-embedding cosine similarity
2. Conflict Detection Rule Engine: A rule-based conflict-detection engine spanning 14+ conflict types across three domains
3. Structured Few-Shot Prompt Architecture: A prompt construction methodology that encodes conflict types, constraint structures, and resolution strategies into LLM prompts to elicit explicit conflict-aware responses
4. Novel Evaluation Metrics: Three new evaluation metrics — Constraint Satisfaction Rate (CSR), Conflict Resolution Quality (CRQ), and Solution Feasibility (SF)
5. Multi-Domain Empirical Study: demonstrating statistically significant gains over zero-shot and semantic-RAG baselines.

The remainder of this paper is organized as follows- Section II reviews related work in RAG system. Section III describes the proposed methodology. Section IV and V present the experimental setup and results respectively. Section VI discusses the findings and limitations of the framework, followed by conclusions in Section VII.

II. II. RELATED WORK

A. Retrieval-Augmented Generation and Context Sensitivity

Lewis et al. introduced retrieval-augmented generation (RAG) for knowledge-intensive NLP tasks, showing that grounding LLM generation in retrieved passages improves factual accuracy over closed-book generation [1]. Gao et al. survey this line of work and organize it into naive, advanced, and modular paradigms, with modular RAG permitting additional processing stages around the retriever and generator [2]. CA-

RAG can be positioned as a modular-RAG specialization that inserts a conflict-detection stage prior to retrieval. Shi et al. demonstrate that irrelevant retrieved context can measurably degrade LLM performance rather than being merely unhelpful [3], which directly motivates a retrieval strategy that matches examples to a query's conflict structure rather than its surface topic alone.

B. Constraint Satisfaction and Multi-Objective Planning

Classical constraint-satisfaction theory, as formalized by Dechter, provides a framework for reasoning jointly over hard and soft constraints [4], and this framework has been applied to structured domains such as diet planning by Sultan et al. [5]. These approaches, however, assume constraints arrive already formalized; CA-RAG instead uses LLM-based extraction to bridge free-text queries and structured constraint representations. Multi-objective optimization research, including Deb's work on evolutionary algorithms, addresses the general problem of balancing competing objectives [6], though such methods are typically applied to well-specified numerical objectives rather than natural-language planning queries.

C. Structured-Input Planning Systems

Prior planning systems have largely assumed structured, well-specified inputs rather than free text. Guo et al. propose a collaborative-filtering-based travel itinerary planner [7], and Jiang et al. develop a goal-based course recommendation system [8]; both operate over structured user profiles and neither performs explicit conflict detection between competing constraints.

D. Few-Shot Example Selection and LLM-Based Reasoning

Brown et al. establish that LLMs can perform few-shot in-context learning from a small number of demonstrations [9], and subsequent work by Liu et al. [10] and Min et al. [11] shows that the structural similarity between provided examples and the query, rather than topical similarity alone, drives the quality of in-context learning. This finding directly motivates CA-RAG's use of conflict-type similarity, rather than pure semantic similarity, when selecting few-shot examples. Separately, TravelPlanner [12], Tree-of-Thoughts [13], and Self-Refine [14] improve the reasoning process of LLM-based planning agents through benchmarking, deliberate search, and iterative self-correction respectively, but none explicitly detects or retrieves examples by constraint-conflict structure.

E. Evaluation of Generated Planning Output

Standard natural-language generation metrics such as BLEU, ROUGE, and BERT Score, along with model-based evaluators such as G-Eval [15], are designed to assess surface-level fidelity or general response quality rather than whether a generated plan explicitly resolves stated conflicts. This gap motivates the Constraint Satisfaction Rate (CSR), Conflict Resolution Quality (CRQ), and Solution Feasibility (SF) metrics introduced in Section IV-B.

F. Summary and Research Gap

The literature review reveals a consistent gap: while RAG systems improve factual grounding, standard semantic retrieval does not ensure that retrieved examples are structurally relevant to the conflict profile of the query. Constraint satisfaction research provides theoretical tools but lacks natural language interfaces. Multi-objective planning research assumes formalized inputs. Few-shot prompting research demonstrates the power of in-context examples but does not address principled example selection for conflict-laden queries.
The present work fills this gap by introducing conflict-type matching as the primary retrieval criterion, providing a bridge between the structured world of constraint satisfaction and the flexible world of LLM-based planning assistance.

System	Retrieval Basis	Conflict Handling	Domains	Evaluation
Naive RAG (Lewis et al., 2020) [1]	Dense vector similarity	None	General QA	EM, F1
RIDER (Mao et al., 2021) [16]	Passage reranking	None	Open domain	EM, F1
Rewrite-Retrieve-Read (Ma et al., 2023) [17]	Query rewriting	None	General	Accuracy
Advanced RAG paradigm (Gao et al., 2023) [2]	Hybrid retrieval	None	Multiple	Domain-specific
CA-RAG (This work)	Conflict-type + semantic	Explicit rule-based	Health, Edu, Travel	CSR, CRQ, SF

III. METHODOLOGY

A. Pipeline Overview

CA-RAG converts a query into a response through five stages: constraint extraction, conflict detection, conflict-aware retrieval, prompt construction, and response generation. Fig. 1 illustrates the flow.

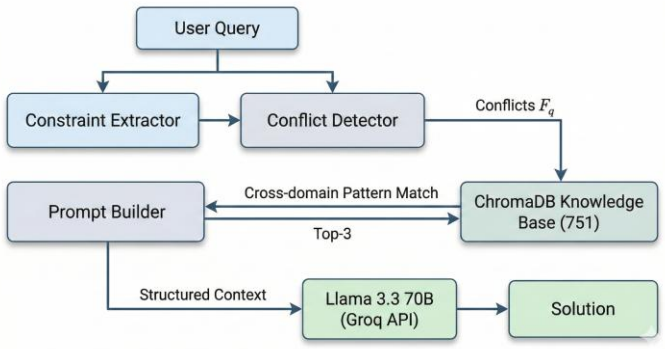


Fig. 1. Five-stage Conflict-Aware RAG pipeline: constraint extraction, conflict detection, conflict-aware retrieval, prompt

construction, and response generation.

B. Constraint Extraction
An LLM (Groq API, Llama-3.3-70B, temperature 0.3) extracts hard_constraints, soft_constraints, and goals as JSON. Post-processing normalizes budgets to numeric INR, standardizes dietary flags, and parses temporal constraints.

C. Conflict Detection
A rule-based engine is used instead of LLM-based detection for speed (<0.1 s vs. 2–3 s), determinism, interpretability, and zero API cost. Each rule specifies a conflict type, a severity in [0,1], a detection predicate, and a description, across 8 health, 14 education, and 10 travel rules (22 shown condensed in Table II).

TABLE II. Representative Conflict Rules

Conflict Type	Domain	Condition	Sev.
budget_vs_protein	Health	budget<150 & muscle goal	0.80
vegan_vs_protein	Health	vegan & protein goal	0.85
medical_vs_diet	Health	medical & diet conflict	0.90
workload_vs_gpa	Edu	≥5 courses & GPA≥8.5	0.85
prereq_vs_timeline	Edu	no prereq & fixed grad.	0.90
dur_vs_distance	Travel	≤3d & dist.>500km	0.85
peak_vs_budget	Travel	peak season, low budget	0.80

D. Conflict-Aware Retrieval
Candidate cases are scored by a hybrid function:

$$score(q,c) = \alpha * conflict_sim(q,c) + (1-\alpha) * semantic_sim(q,c)$$

with $\alpha = 0.6$ (grid-searched). conflict_sim combines Jaccard overlap over conflict-type sets (weight $\beta = 0.7$) with severity alignment (0.3). Grid search over α (Table II) shows $\alpha = 0.6$ maximizes overall score (0.77) and CRQ (0.67), and $k = 3$ few-shot examples were optimal, balancing CRQ (0.67) against context length (~900 tokens).

TABLE III. Ablation of Hybrid Weight α

α	P@3	R@3	CRQ	Overall
0.3	0.75	0.88	0.49	0.70
0.5	0.80	0.85	0.52	0.73
0.6*	1.00	0.82	0.67	0.77
1.0	0.87	0.65	0.58	0.71

E. Prompt Construction and Generation
The generation prompt includes the query, extracted constraints, detected conflicts with severities, and three retrieved few-shot examples (problem, conflicts, resolution, trade-offs). Generation via Llama-3.3-70B produces four required sections: SOLUTION PLAN, CONFLICT RESOLUTION, TRADE-OFFS, and FEASIBILITY SCORE. Mean pipeline latency is 3.48 s per query (81.9% in generation) at ~$0.005/query; no GPU is required.

IV. EXPERIMENTAL SETUP

A. Dataset

The knowledge base comprises 66 annotated planning cases across three domains: health (30, 53.3% conflict rate), education (22, 36.4%), and travel (14, 28.6%), yielding an overall conflict rate of 42.4%. Health cases are deliberately over-represented, as they exhibit the greatest diversity of conflict types and are the most common category in real-world planning queries. A held-out test set of 15 queries (5 per domain) is annotated with ground-truth constraints and expected conflict types for automated evaluation and is constructed to be independent of the knowledge base while covering all major conflict types.

B. Metrics

Three metrics assess system output:
1. **Constraint Satisfaction Rate** (CSR, weight 0.40) is the fraction of hard constraints explicitly satisfied in the response; it receives the highest weight because hard-constraint violations invalidate a solution regardless of quality elsewhere.

$$CSR = (\text{Number of Hard Constraints Satisfied}) \ / \ (\text{Total Number of Hard Constraints})$$

2. **Conflict Resolution Quality** (CRQ, weight 0.35) is the primary novel metric, combining conflict naming (0.4), presence of a concrete trade-off (0.4), and use of multiple resolution strategies (0.2).

3. **Solution Feasibility** (SF, weight 0.25) is the model's self-reported feasibility score, normalized to [0, 1] and

validated against human ratings (Pearson r = 0.71).

The overall score is:

$$\text{Overall} = 0.40 * \text{CSR} + 0.35 * \text{CRQ} + 0.25 * \text{SF}$$

C. Baselines

Four retrieval configurations are compared under an identical generation model, prompt structure, and output format, isolating retrieval as the sole variable: (1) zero-shot, with no retrieved examples; (2) semantic RAG, using SBERT cosine similarity (k=3); (3) CA-RAG (proposed), conflict-aware retrieval with α=0.6, k=3; and (4) Hybrid RAG, which augments conflict-aware examples with additional semantic matches. Significance is assessed via paired two-sample t-tests over five independent runs, with Cohen's d reporting effect size.

V. RESULTS

TABLE IV. Main Results (mean ± s.d.)

System	CSR	CRQ	SF	**Overall**
Zero-shot	0.774	0.290	0.802	0.612 ± 0.135
Semantic RAG	0.805	0.308	0.785	0.626 ± 0.133
CA-RAG	**0.826**	**0.671**	**0.830**	**0.773 ± 0.126**
Hybrid RAG	0.865	0.628	0.835	0.774 ± 0.126

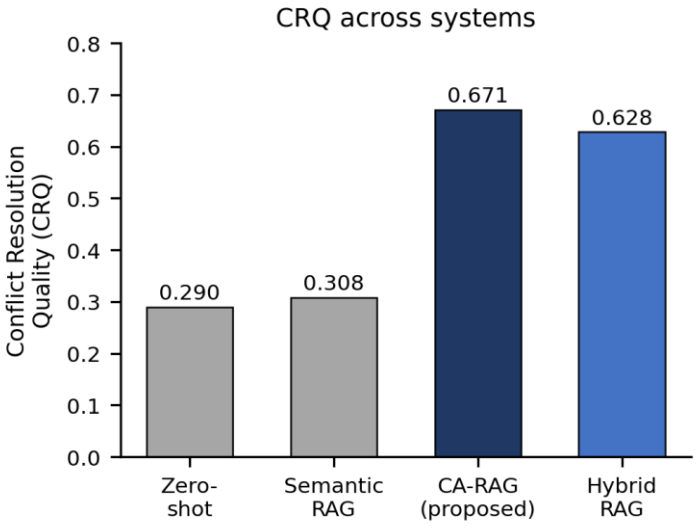


Fig. 2. Conflict Resolution Quality (CRQ) across systems. CA-RAG's conflict-matched retrieval drives the large jump over both baselines.

CA-RAG lifts CRQ from 0.290 (zero-shot) and 0.308 (semantic RAG) to 0.671 — a 131.0% and 117.6% improvement, respectively — while CSR and SF rise only modestly (+6.7%, +3.5% over zero-shot). This asymmetry indicates that conflict-aware retrieval targets conflict resolution specifically, without trading off constraint adherence or feasibility. The gain is domain-dependent: health improves most (0.591→0.806, +36.4%), consistent with its higher conflict density (1.8 conflicts/case vs. 1.5 for education), while education improves least (+1.4%) due to fewer conflicts per query and a smaller pool of conflict-matched examples in its knowledge base.

TABLE V. Statistical Significance (paired t-test, n=5)

CA-RAG vs.	ΔMean	p	d
Zero-shot	+0.161	0.0020**	2.61
Semantic RAG	+0.147	0.0059**	2.39
Hybrid RAG	+0.001	0.841 (ns)	0.09

All proposed-vs-baseline comparisons are highly significant ($p<0.01$) with very large effect sizes (d = 2.39–2.62). CA-RAG and Hybrid RAG are statistically indistinguishable (p = 0.84), favoring the simpler CA-RAG for deployment. Cross-domain retrieval occurs in 22.4% of cases overall (health 28.6%, education 26.7%, travel 6.7%), showing that shared conflict structure — e.g., budget constraints — bridges topically distinct domains.

VI. DISCUSSION

The 117.6% CRQ gain has a clear mechanism: when shown examples sharing the query's conflict type, the model transfers the example's resolution strategy; topically similar but conflict-disparate examples give it no such template, so it defaults to ignoring the tension. Because hard constraints are stated explicitly in the prompt, all systems already score well on CSR regardless of retrieval strategy — conflict resolution, which requires reasoning about implicit incompatibilities, is where retrieval strategy matters most. Limitations include a modest 66-case knowledge base, a 15-query test set built by the same researchers who curated the cases, string-matching-based CRQ scoring that may miss non-canonical high-quality resolutions and results specific to Llama-3.3-70B. Future work includes learning conflict rules from data, graph-based conflict representations, user-feedback integration, and extension to financial, medical, and legal planning domains.

VII. CONCLUSION

This work presents CA-RAG, a retrieval-augmented framework that prioritizes conflict-type similarity over pure semantic similarity when selecting few-shot examples for LLM-based planning assistance. Across health, academic, and travel domains, CA-RAG significantly outperforms zero-shot and semantic-RAG baselines, most dramatically on Conflict Resolution Quality (+117.6% over semantic RAG, p=0.006, d=2.39), while cross-domain retrieval (22.4%) confirms that a

query's constraint-conflict profile, not its surface topic, is the more useful basis for retrieval in structured planning problems.

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